Canonical Physics and Conventions Reference for `cqed_sim`

Post-cleanup and protocol-alignment reference

March 10, 2026

This document is the source of truth for the reusable simulator code under `cqed_sim/`. It records the active physics conventions, the low-level state-preparation, measurement, and spectral primitives that remain inside the library, the cleaned package boundary versus `examples/`, and the tests that directly back the main conventions.

1 Overview of cqed_sim

`cqed_sim/` now contains only reusable simulator code:

- `core`: Hilbert-space conventions, frames, the universal subsystem-based cQED model layer, the two-mode and three-mode compatibility wrappers, reusable state-preparation helpers, static-spectrum helpers, ideal gates, and manifold-frequency helpers.
- `pulses`, `sequence`, `sim`: pulse objects, envelopes, channel compilation, hardware distortion, the solver path, collapse operators, and extractors.
- `measurement`: reusable qubit readout abstractions, physical readout-chain models, and measurement utilities that remain backend-agnostic and library-oriented.
- `calibration`, `observables`, `operators`, `tomo`, `io`, `plotting`, `unitary_synthesis`: reusable helpers that still belong to the library surface.

Study, workflow, paper-reproduction, audit, and smoke-test code remains under `examples/`. The alignment note for the experiment-facing cleanup is kept in `physics_and_conventions/experimental_protocol_a`

2 State Preparation and Measurement Primitives

The reusable preparation and measurement helpers that remain in the library are implemented in:

- `cqed_sim/core/state_prep.py`: `StatePreparationSpec`, reusable subsystem state specs, and helpers such as `prepare_state(...)`.
- `cqed_sim/measurement/qubit.py`: `QubitMeasurementSpec` and `measure_qubit(...)`.
- `cqed_sim/measurement/readout_chain.py`: `ReadoutResonator`, `PurcellFilter`, `AmplifierChain`, and `ReadoutChain`.
- `cqed_sim/sim/noise.py`: reusable noise helpers such as `pure_dephasing_time_from_t1_t2(...)` used by library code and example workflows.

High-level experiment recipes and canned workflows no longer live under `cqed_sim/`; they now live under `examples/workflows/`. These library objects are not a second solver stack. They are reusable primitives that compose with `SequenceCompiler` plus `simulate_sequence(...)`. In particular:

- State preparation may return either a ket or a density matrix. If any subsystem spec is mixed, the joint initial state is promoted to a density matrix before tensoring.
- `measure_qubit(...)` returns exact probabilities by default. Sampled outcomes appear only when `shots` is set.
- When a confusion matrix is supplied, the code applies $\mathbf{p}_{\text{obs}} = M\mathbf{p}_{\text{latent}}$ with columns ordered as latent (g, e) and rows ordered as reported (g, e).
- If `QubitMeasurementSpec.readout_chain` is omitted, synthetic I/Q points are still generated from the previous Gaussian cluster model.
- If a `ReadoutChain` is provided, the measurement layer instead uses a resonator response, optional Purcell filter, amplifier noise, and mixer-style I/Q integration. The same API can also report γ_{meas} and a Purcell-limited T_1, and can optionally apply readout-induced dephasing plus Purcell relaxation as a reduced-state backaction model.

Direct tests: `tests/test_26_state_prep_and_measurement.py`, `tests/experiment/test_readout_chain.py`.

3 Physical Readout Chain

The physical readout-chain path models a driven dispersive resonator with optional Purcell filtering and additive amplifier noise:

- `ReadoutResonator` stores ω_r, κ, the qubit-resonator coupling g, the complex drive amplitude ϵ, and an optional runtime-convention dispersive pull χ.
- The steady-state pointer amplitudes are

$$\alpha_{g/e}^{(\text{ss})} = \frac{-i\epsilon}{\kappa_{\text{eff}}/2 + i(\omega_{r,g/e} - \omega_d)},$$

  with $\omega_{r,g} = \omega_r$ and $\omega_{r,e} = \omega_r + \chi$ in the runtime projector convention.
- The implemented measurement-induced dephasing rate is

$$\gamma_{\text{meas}} = \frac{\kappa_{\text{eff}}}{2}|\alpha_e - \alpha_g|^2,$$

  which reproduces the expected small-χ scaling $\gamma_{\text{meas}} \propto \chi^2\bar{n}/\kappa$.
- The Purcell filter follows the standard frequency-dependent linewidth picture:

$$\kappa_{\text{eff}}(\omega) = \frac{4J^2\kappa_f}{\kappa_f^2 + 4(\omega - \omega_f)^2},$$

  so the Purcell rate is evaluated as $\Gamma_P = (g/\Delta)^2\kappa_{\text{eff}}(\omega_q)$.
- `AmplifierChain` applies linear gain and additive thermal noise before mixer-phase rotation and I/Q integration.

Direct tests: `tests/experiment/test_readout_chain.py`.

4 Sequential Sideband Reset Approximation

The sequential storage-reset workflow now lives in `examples/workflows/sequential_sideband_reset.py`. It remains an example-level orchestration built on the existing explicit three-mode dispersive runtime rather than a second simulator stack. The modeled protocol is:

$$|g, 0_r, n_s\rangle \overset{\text{storage red sideband}}{\leftrightarrow} |f, 0_r, n_s - 1\rangle \overset{\text{readout red sideband}}{\leftrightarrow} |g, 1_r, n_s - 1\rangle \xrightarrow{\kappa_r} |g, 0_r, n_s - 1\rangle.$$

For initial $|e, 0_r, n_s\rangle$ population, the helper first applies an explicit $e \to f$ transmon pulse and then performs the readout-sideband dump before entering the photon-removal loop.

This is a controlled effective-model approximation. The readout-assisted evacuation stage is represented as a driven effective $g \leftrightarrow f$ red sideband on the readout mode followed by explicit Lindblad readout loss, not as an autonomous always-on microscopic transmon–readout exchange Hamiltonian.

5 Static Spectrum Convention

The reusable spectral helper in `cqed_sim/core/spectrum.py` diagonalizes the static Hamiltonian selected by the caller and returns a structured `EnergySpectrum` result with both raw eigenvalues and vacuum-referenced energies.

- The diagonalized operator is exactly `model.static_hamiltonian(frame)` for the chosen `FrameSpec`.
- Reported `EnergySpectrum.energies` are shifted by subtracting the energy of the bare vacuum basis state.
- The bare vacuum means every subsystem is in level 0: $|g, 0\rangle$ for the two-mode model, $|g, 0_s, 0_r\rangle$ for the three-mode model, and $|0\rangle$ for cavity-only models.
- This shift changes only the reported zero of energy. It does not change which Hamiltonian was diagonalized or which transition frequencies follow from energy differences.
- In the lab frame, the low-lying spectrum is the most directly interpretable dressed ladder. In rotating frames, excited levels can appear at negative energy even though the vacuum remains the zero-reference state.

Direct tests: `tests/analysis/test_spectrum.py`.

6 Hilbert Space Structure

Two-mode runtime

- Runtime ordering is qubit first, storage second:

$$|q, n\rangle = |q\rangle_{\text{qubit}} \otimes |n\rangle_{\text{storage}}.$$

- The flattened basis index is $\texttt{index}(q, n) = q\, n_{\text{cav}} + n$, implemented in `cqed_sim/core/conventions.py`.
- Computational basis is $|g\rangle = |0\rangle$, $|e\rangle = |1\rangle$.

Three-mode runtime

- The three-mode extension uses

$$|q, n_s, n_r\rangle = |q\rangle_{\text{qubit}} \otimes |n_s\rangle_{\text{storage}} \otimes |n_r\rangle_{\text{readout}}.$$

- The flattened basis index is

$$\texttt{index}(q, n_s, n_r) = ((q\, n_{\text{storage}}) + n_s)\, n_{\text{readout}} + n_r.$$

- This ordering is implemented in `cqed_sim/core/conventions.py` and `cqed_sim/core/readout_model.py`, and all new extractors assume subsystem 0 is the transmon, subsystem 1 is storage, subsystem 2 is readout.

Universal model layer

- The reusable model core now lives in `cqed_sim/core/universal_model.py`.
- `UniversalCQEDModel` accepts an optional `TransmonModeSpec`, a sequence of `BosonicModeSpec` entries, explicit `DispersiveCouplingSpec` entries, and the existing generic `CrossKerrSpec`, `SelfKerrSpec`, and `ExchangeSpec` terms.
- Tensor ordering remains explicit: the transmon is first when present, followed by bosonic modes in declaration order.
- `DispersiveTransmonCavityModel` and `DispersiveReadoutTransmonStorageModel` preserve their previous public constructors and delegate their Hamiltonian and operator logic to that shared universal layer.

Synthesis subspace

The unitary-synthesis layer retains the same full-space qubit-first ordering, but restricted subspace bases are explicitly reordered in `cqed_sim/unitary_synthesis/subspace.py`; for example, a qubit-cavity block is ordered as $|g, 0\rangle, |e, 0\rangle, |g, 1\rangle, |e, 1\rangle, \ldots$.

Direct tests: `tests/test_25_tensor_product_convention.py`, `tests/test_27_three_mode_model.py`, `tests/unitary_synthesis/test_subspace.py`.

7 Operator Conventions

- The simulator uses QuTiP Pauli matrices without sign changes. Therefore

$$\sigma_z|g\rangle = +|g\rangle, \qquad \sigma_z|e\rangle = -|e\rangle.$$

- The core Hamiltonians use the excitation number operator $n_q = b^\dagger b$, not $\sigma_z/2$. For two levels,

$$n_q = |e\rangle\langle e| = \frac{I - \sigma_z}{2}.$$

- The shared operator builder now lives in `cqed_sim/core/universal_model.py`; the specialized wrapper models keep the same user-facing operator semantics while delegating tensor lifting and caching to that shared path.

- Two-mode operators remain embedded as $a = I_q \otimes a_{\text{storage}}$ and $b = b_{\text{transmon}} \otimes I_s$.
- Three-mode operators remain embedded as

$$b = b_{\text{transmon}} \otimes I_s \otimes I_r, \quad a_s = I_q \otimes a_{\text{storage}} \otimes I_r, \quad a_r = I_q \otimes I_s \otimes a_{\text{readout}}.$$

- The generic runtime now resolves drive operators through `model.drive_coupling_operators()`, so the solver no longer hard-codes a qubit-plus-one-mode assumption.

Direct tests: `tests/test_16_ideal_primitives_and_extractors.py`, `tests/test_20_gaussian_iq_convention` `tests/test_25_tensor_product_convention.py`, `tests/test_27_three_mode_model.py`.

8 Hamiltonian Definitions

Two-mode runtime Hamiltonian

The reusable two-mode Hamiltonian exposed by `cqed_sim/core/model.py` and assembled by the shared universal model layer is

$$H_0 = \delta_c n_c + \delta_q n_q + \frac{\alpha}{2} b^{\dagger 2} b^2 + K(n_c) + \Xi(n_c) n_q,$$

with

$$K(n_c) = \frac{\texttt{kerr}}{2} n_c(n_c - 1) + \cdots, \qquad \Xi(n_c) = \texttt{chi}\, n_c + \texttt{chi2}\, n_c(n_c-1) + \texttt{chi3}\, n_c(n_c-1)(n_c-2) + \cdots.$$

`cqed_sim/core/frequencies.py` makes the intended transition law explicit:

$$\omega_{ge}(n) = \omega_{ge}(0) + \Xi(n).$$

Textbook-to-code translation

With the active runtime convention,

$$n_q = \frac{I - \sigma_z}{2}.$$

Therefore

$$\delta_q n_q = \frac{\delta_q}{2} I - \frac{\delta_q}{2} \sigma_z, \qquad \Xi(n_c) n_q = \frac{\Xi(n_c)}{2} I - \frac{\Xi(n_c)}{2} \sigma_z.$$

Up to identity and storage-only shifts,

$$H_{q,\text{runtime}} \doteq -\frac{1}{2}\big(\delta_q + \Xi(n_c)\big)\sigma_z.$$

If one instead uses the textbook sign

$$\sigma_z^{\text{text}} = |e\rangle\langle e| - |g\rangle\langle g| = -\sigma_z,$$

then

$$H_{q,\text{runtime}} \doteq \frac{1}{2}\big(\delta_q + \Xi(n_c)\big)\sigma_z^{\text{text}}.$$

Hence negative runtime `chi` lowers the $|g, n\rangle \leftrightarrow |e, n\rangle$ transition frequency, and positive runtime `chi` raises it.

Comparison with the synthesis drift convention

`cqed_sim/unitary_synthesis/sequence.py` now uses the same projector-based dispersive and Kerr semantics as the runtime model:

$$E_{g,n} = \delta_c n + K(n), \qquad E_{e,n} = \delta_c n + \delta_q + \Xi_{\text{synth}}(n) + K(n).$$

Its transition shift is therefore

$$E_{e,n} - E_{g,n} = \delta_q + \Xi_{\text{synth}}(n).$$

Matching the runtime shift $\delta_q + \Xi_{\text{runtime}}(n)$ gives

$$\Xi_{\text{synth}}(n) = \Xi_{\text{runtime}}(n),$$

so componentwise

$$\chi_{\text{synth}} = \chi_{\text{runtime}}, \qquad \chi_{2,\text{synth}} = \chi_{2,\text{runtime}}, \qquad \chi_{3,\text{synth}} = \chi_{3,\text{runtime}}.$$

Likewise the cavity self-Kerr contribution is now shared directly:

$$K_{\text{synth}}(n) = K_{\text{runtime}}(n).$$

The compatibility fields `synthesis_chi` and `synthesis_chi_2` in `cqed_sim/analysis/parameter_translation.py` are therefore aligned copies rather than sign-converted quantities.

Kerr free-evolution workflow presets

The example workflow in `examples/workflows/kerr_free_evolution.py` builds a two-mode storage–transmon model from human-readable frequencies in Hz and converts them to the runtime's angular-frequency units with `physics_and_conventions.conventions.to_internal_units()`. Its default frame is

$$\omega_{c,\text{frame}} = \omega_c, \qquad \omega_{q,\text{frame}} = \omega_q,$$

so the undriven evolution isolates the nonlinear Kerr and dispersive terms after removing the bare storage and qubit carrier rotations.

The Hamiltonian used by this workflow is therefore exactly the two-mode runtime Hamiltonian above, evaluated in that rotating frame:

$$\frac{H_{\text{free}}}{\hbar} = \frac{\alpha}{2} b^{\dagger 2} b^2 + \frac{K}{2} n_c(n_c - 1) + \frac{K'}{6} n_c(n_c - 1)(n_c - 2) + \left[\chi n_c + \chi' n_c(n_c - 1) + \chi'' n_c(n_c - 1)(n_c - 2)\right] n_q.$$

When the qubit is initialized in $|g\rangle$, the reduced cavity state undergoes pure self-Kerr evolution; the dispersive coefficients remain part of the joint Hamiltonian and matter whenever a different qubit state preparation is used.

The workflow ships with two named parameter sets:

- `phase_evolution`: $\omega_q/2\pi = 7627.05\,\text{MHz}$, $\omega_c/2\pi = 8226.787\,\text{MHz}$, $\omega_{ro}/2\pi = 9376.75\,\text{MHz}$, $\alpha_q/2\pi = 283.4\,\text{MHz}$, $K/2\pi = -107.9\,\text{kHz}$, $K'/2\pi = 3.4\,\text{kHz}$, $\chi/2\pi = -8281.3\,\text{kHz}$, $\chi'/2\pi = 48.8\,\text{kHz}$, $\chi''/2\pi = 0.5\,\text{kHz}$.
- `value_2`: same ω_q, ω_c, ω_{ro}, and α_q, with $K/2\pi = -106.2\,\text{kHz}$, $K'/2\pi = 3.4\,\text{kHz}$, $\chi/2\pi = -8273\,\text{kHz}$, $\chi'/2\pi = 60.9\,\text{kHz}$, and χ'' taken as zero because no alternate value was supplied.

The readout frequency ω_{ro} is stored with the preset metadata for consistency with the measured parameter set, but it is not part of the two-mode Kerr free-evolution Hamiltonian.

Three-mode storage-transmon-readout Hamiltonian

The three-mode model exposed by `cqed_sim/core/readout_model.py` and assembled by the shared universal model layer implements

$$\frac{H}{\hbar} = \delta_s n_s + \delta_r n_r + \delta_q n_q + \frac{\alpha}{2} b^{\dagger 2} b^2 + \chi_s n_s n_q + \chi_r n_r n_q + \chi_{sr} n_s n_r + \frac{K_s}{2} n_s(n_s - 1) + \frac{K_r}{2} n_r(n_r - 1).$$

This is intentionally kept in the same runtime convention style as the two-mode path:

- $n_q = b^\dagger b$ is the transmon excitation projector/number operator.
- Positive χ_s or χ_r raises the transmon transition frequency as storage or readout occupancy increases.
- Positive χ_{sr} raises the storage frequency with readout occupancy and raises the readout frequency with storage occupancy.
- Positive K_s or K_r raises the adjacent-mode spacing by $+Kn$.

The model also exposes analytic helpers for the basis energies and the qubit, storage, and readout transition frequencies.

Optional nonlinear and exchange terms

`cqed_sim/sim/couplings.py` now provides explicit builder functions for

$$\chi_{ab} a^\dagger a\, b^\dagger b, \qquad +\frac{K}{2} a^{\dagger 2} a^2, \qquad J(a^\dagger b + a b^\dagger),$$

and the model classes accept `CrossKerrSpec`, `SelfKerrSpec`, and `ExchangeSpec` entries that are injected into the static Hamiltonian assembly. This extends the existing dispersive models without replacing their original API.

Canonical parameter-definition table

Name	Code-level meaning	Algebra used	Caveat
`chi`	two-mode first-order dispersive pull	$+$`chi` $n_c n_q$, $\omega_{ge}(n) = \omega_{ge}(0) + \chi n$	negative values lower ω_{ge} with storage photons
`chi2`	second-order pull, exposed by `chi_higher[0]`	$+\chi_2 n_c(n_c - 1) n_q$	falling-factorial coefficient
`chi3`	third-order pull, exposed by `chi_higher[1]`	$+\chi_3 n_c(n_c - 1)(n_c - 2) n_q$	falling-factorial coefficient
`chi_s`	storage-transmon dispersive pull in the three-mode model	$+\chi_s n_s n_q$	negative values lower ω_{ge} with storage photons
`chi_r`	readout-transmon dispersive pull in the three-mode model	$+\chi_r n_r n_q$	negative values lower ω_{ge} with readout photons

Name	Code-level meaning	Algebra used	Caveat
`chi_sr`	storage-readout cross-Kerr	$+\chi_{sr} n_s n_r$	positive values raise both bosonic mode frequencies with occupancy of the other mode
`kerr`	two-mode storage self-Kerr coefficient	$+\texttt{kerr}\, n_c(n_c-1)/2$	positive values raise the adjacent storage spacing by $+\texttt{kerr}\, n$
higher Kerr	tuple `kerr_higher`	$K_m n_c^{\underline{m}}/m!$	falling-factorial hierarchy
`kerr_s`	three-mode storage self-Kerr	$+\texttt{kerr_s}\, n_s(n_s-1)/2$	same sign convention as `kerr`
`kerr_r`	three-mode readout self-Kerr	$+\texttt{kerr_r}\, n_r(n_r-1)/2$	same sign convention as `kerr`
`alpha`	transmon Duffing anharmonicity	$\texttt{alpha}\, b^{\dagger 2} b^2/2$	inactive for strict two-level truncation
σ_z	active runtime Pauli-Z	$\lvert g\rangle\langle g\rvert - \lvert e\rangle\langle e\rvert = I - 2n_q$	QuTiP standard sign
n_q	transmon excitation operator	$b^\dagger b$; for two levels $n_q = \lvert e\rangle\langle e\rvert$	repo-specific implementation choice versus $\sigma_z/2$ notation
δ_q	qubit residual detuning	$\omega_q - \omega_q^{\text{frame}}$	only defined after choosing a frame
δ_c	two-mode storage residual detuning	$\omega_c - \omega_c^{\text{frame}}$	same frame caveat
δ_s	three-mode storage residual detuning	$\omega_s - \omega_s^{\text{frame}}$	ω_s^{frame} is stored as `omega_c_frame` with an alias
δ_r	three-mode readout residual detuning	$\omega_r - \omega_r^{\text{frame}}$	only used in the three-mode path

Standard-compatible versus repo-specific

- Standard-compatible: qubit-first tensor ordering, QuTiP Pauli signs, rad/s and seconds internally, Duffing/Kerr terms, Lindblad dynamics, and the canonical rule that negative χ lowers the qubit transition with photon number while positive χ raises it.
- Repo-specific: writing the transmon term in terms of n_q, storing higher-order pulls and Kerr terms in falling-factorial form, and using complex envelopes with $e^{+i(\omega t+\phi)}$.
- Needs caution: raw `Pulse.carrier` remains the negative of the rotating-frame transition frequency, so user-facing spectroscopy axes should use `carrier_for_transition_frequency(...)` or `transition_frequency_from_carrier(...)` rather than raw carriers.

9 Frequency and Time Units

- Internal Hamiltonian coefficients and rotating-frame frequencies are in rad/s.
- Pulse durations, timeline steps, and solver `max_step` values are in seconds.
- The repository now centralizes conversion helpers in `physics_and_conventions/conventions.py`, including `to_internal_units(...)` and `from_internal_units(...)`.

- Example-side configuration files may still carry `_hz`, `_ns`, or `_1_per_s` field names, but those values are converted before constructing library models or noise objects.

Direct tests: `tests/test_05_detuning_and_frames.py`, `tests/test_10_chi_convention.py`, `tests/test_13_ha` `tests/test_27_three_mode_model.py`.

10 Parameter Translation, Calibration Targets, and Backends

- `cqed_sim/analysis/parameter_translation.py` now provides `from_transmon_params(...)` and `from_measured(...)`. Bare transmon parameters use the usual large-E_J/E_C expansion $\omega_{01} \approx \sqrt{8E_JE_C} - E_C$ and $\alpha \approx -E_C$, while dressed χ and χ_2 are extracted by matching a low-dimensional Duffing-plus-resonator diagonalization to the runtime form $\omega_{ge}(n) = \omega_q + \chi n + \chi_2 n(n-1)$.
- `cqed_sim/calibration_targets/` now provides reusable spectroscopy, Rabi, Ramsey, T_1, $T_{2,\mathrm{echo}}$, and DRAG-tuning surrogate targets that return fitted parameters, uncertainties, and raw simulated data.
- `cqed_sim/backends/` now provides `NumPyBackend` and `JaxBackend` plus a dense piecewise-constant solver path in `cqed_sim/sim/solver.py`. The default runtime still uses QuTiP; the backend path is opt-in through `SimulationConfig.backend`.

Direct tests: `tests/analysis/test_parameter_translation.py`, `tests/calibration_targets/test_targets.p` `tests/sim/test_backends.py`.

11 Rotating Frame Conventions

- Frames are represented by `FrameSpec(omega_c_frame, omega_q_frame, omega_r_frame)` in `cqed_sim/core/frame.py`.
- The legacy field `omega_c_frame` remains the storage/cavity frame frequency. The readout model adds the alias property `omega_s_frame` for clarity.
- The two-mode path uses $\delta_c = \omega_c - \omega_c^{\mathrm{frame}}$ and $\delta_q = \omega_q - \omega_q^{\mathrm{frame}}$.
- The three-mode path uses $\delta_s = \omega_s - \omega_s^{\mathrm{frame}}$, $\delta_r = \omega_r - \omega_r^{\mathrm{frame}}$, and $\delta_q = \omega_q - \omega_q^{\mathrm{frame}}$.
- `manifold_transition_frequency(...)` still refers only to the two-mode path; the three-mode model instead exposes `qubit_transition_frequency(...)`, `storage_transition_frequency(...)`, and `readout_transition_frequency(...)`.

Direct tests: `tests/test_05_detuning_and_frames.py`, `tests/test_21_qubox_convention_reconciliation.py` `tests/test_27_three_mode_model.py`.

12 Drive Hamiltonians

- `cqed_sim/pulses/pulse.py` samples analytic or discrete envelopes as

$$\epsilon(t) = \texttt{amp}\,\texttt{envelope}(t)\, e^{i(\texttt{carrier}t+\texttt{phase})}.$$

- `cqed_sim/sim/runner.py` inserts $\epsilon(t)$ and $\epsilon^*(t)$ as raising and lowering couplings obtained from the model, rather than from a hard-coded target list.
- Because the runtime drive term is $\epsilon(t)\, b^\dagger + \epsilon^*(t)\, b$, a rotating-frame transition at frequency $\omega_{\text{transition}}$ is resonantly addressed by `carrier` $= -\omega_{\text{transition}}$. The public helpers `carrier_for_transition_freq` and `transition_frequency_from_carrier(...)` make that mapping explicit.
- User-facing spectroscopy examples and the updated `examples/workflows/cqed_sim_usage_examples.ipynb` therefore plot physical transition detuning on the x-axis and convert that value to the internal raw carrier with `carrier_for_transition_frequency(...)` before simulation.
- The two-mode path supports `"cavity"` or `"storage"`, `"qubit"`, and the legacy `"sideband"` target, plus the structured targets `TransmonTransitionDriveSpec(lower_level, upper_level)` and `SidebandDriveSpec(mode, lower_level, upper_level, sideband)`.
- The three-mode path supports `"storage"`, `"readout"`, `"qubit"`, and `"transmon"` targets, plus the same structured transition and sideband targets. The readout and storage drives use the same complex-envelope sign convention as the existing qubit and storage drives.
- The effective three-mode drive Hamiltonian is

$$H_d/\hbar = \epsilon_s(t)a_s^\dagger + \epsilon_s^*(t)a_s + \epsilon_q(t)b^\dagger + \epsilon_q^*(t)b + \epsilon_r(t)a_r^\dagger + \epsilon_r^*(t)a_r.$$

Direct tests: `tests/test_20_gaussian_iq_convention.py`, `tests/test_23_sqr_additive_amplitude_correcti` `tests/test_28_three_mode_drives_and_noise.py`, `tests/test_31_multilevel_sideband_extension.py`.

Effective multilevel sideband model

The reusable sideband path added in `cqed_sim/core/drive_targets.py` and `cqed_sim/pulses/builders.py` is an effective rotating-wave Hamiltonian, not a microscopic flux-drive derivation. For a red sideband on a bosonic mode a_m and a selected transmon transition $|\ell\rangle \leftrightarrow |u\rangle$, the implemented operator pair is

$$H_{\text{sb}}/\hbar = \epsilon(t)\, |u\rangle\langle\ell|\, a_m + \epsilon^*(t)\, a_m^\dagger |\ell\rangle\langle u|,$$

with an analogous blue-sideband variant that swaps a_m and $a_m^\dagger$. The public `SidebandDriveSpec` selects the bosonic mode, the ancilla ladder transition, and the red/blue branch. The helper `effective_sideband_rabi_frequency(g, δ)` uses the standard effective-model rule

$$\Omega_{\text{eff}} = \sqrt{(2g)^2 + \delta^2}.$$

This path is intended for fast-control and synchronization studies where the effective manifold dynamics are the quantity of interest.

Kerr-sign notebook diagnosis

The paper-comparison notebook `test_against_papers/prl115_137002_bosonic_controls.ipynb` was audited against the runtime implementation. The core simulator and the documented physics convention both use

$$H_K/\hbar = +\frac{K}{2}a^{\dagger 2}a^2.$$

For the measured parameter set used in the notebook, the runtime evolution matches the documented negative-K behavior. The apparent sign flip comes from interpreting the rotating-frame

coherent-state phase trajectory, not from a flipped self-Kerr term in the Hamiltonian builder. This is now guarded by `tests/test_32_kerr_sign_notebook_regression.py` and exposed through `cqed_sim.experiment.verify_kerr_sign(...)`.

13 Dissipation and Lindblad Terms

- `NoiseSpec` in `cqed_sim/sim/noise.py` now supports `t1`, `transmon_t1`, `tphi`, `tphi_storage`, `tphi_readout`, `kappa`, `nth`, `kappa_storage`, `kappa_readout`, `nth_storage`, and `nth_readout`.
- The implemented convention is $\gamma_1 = 1/T_1$ and $\gamma_\phi = 1/(2T_\phi)$.
- If `transmon_t1=(T1_ge, T1_fe, ...)` is supplied, the collapse-operator builder emits explicit ladder decays $\sqrt{1/T_{1,j}}\,|j-1\rangle\langle j|$. If it is not supplied, the older aggregate $\sqrt{\gamma_1}\,b$ path remains available through `t1`.
- Two-level dephasing uses $\sqrt{\gamma_\phi}\,\sigma_z$ with $\sigma_z = I - 2n_q$. Multi-level transmon dephasing uses $\sqrt{\gamma_\phi}\,n_q$.
- Storage and readout pure dephasing use $\sqrt{1/T_{\phi,s}}\,n_s$ and $\sqrt{1/T_{\phi,r}}\,n_r$ respectively, so a storage $|0\rangle\langle 1|$ coherence decays as $e^{-t/T_{\phi,s}}$.
- In the two-mode path, `kappa` and `nth` still drive storage loss and thermal excitation.
- In the three-mode path, storage and readout can decay independently via `kappa_storage` and `kappa_readout`, with separate thermal occupations.
- The helper `pure_dephasing_time_from_t1_t2(...)` uses $1/T_2 = 1/(2T_1) + 1/T_\phi$ and clamps negative inferred $1/T_\phi$ to zero.
- `simulate_sequence(...)` uses `mesolve` when collapse operators are present or when the initial state is already a density matrix; otherwise it uses `sesolve`.

Direct tests: `tests/test_14_dissipation.py`, `tests/test_18_noise_and_sideband.py`, `tests/test_28_three_` `tests/test_31_multilevel_sideband_extension.py`.

14 State Representation

- Runtime inputs may be kets or density matrices.
- The result wrapper stores the final state, optional state history, expectation traces, and the raw QuTiP solver result.
- The generic extractor path now supports two-mode and three-mode states through subsystem-aware partial traces.
- `reduced_cavity_state(...)` remains specific to the two-mode path. The three-mode path uses `reduced_storage_state(...)` and `reduced_readout_state(...)`.
- `measure_qubit(...)` acts on the reduced transmon state obtained from the runtime state convention. The exact probabilities are taken from the reduced density matrix before any optional confusion-matrix application or finite-shot sampling.
- Synthetic I/Q points remain available for lightweight sampling-only usage.

- The new readout-chain path adds optional state-conditioned resonator traces, amplifier-noise I/Q sampling, and a reduced-state backaction model without changing the default no-chain behavior.

Direct tests: `tests/test_16_ideal_primitives_and_extractors.py`, `tests/test_19_fock_tomo.py`, `tests/test_26_state_prep_and_measurement.py`, `tests/test_27_three_mode_model.py`, `tests/test_28_thre

15 Hilbert Space Truncation

- The two-mode model uses `n_cav` for storage truncation and `n_tr` for transmon truncation.
- The three-mode model uses `n_storage`, `n_readout`, and `n_tr`.
- The universal model layer also supports transmon-only and cavity-only special cases through the same subsystem-spec representation, while the wrapper models keep the historically used two-mode and three-mode constructor surfaces.
- The multilevel sideband and structured-transition helpers assume the selected ancilla levels lie within the chosen transmon truncation and will raise if the requested transition falls outside $[0, \texttt{n_tr} - 1]$.
- State preparation helpers pad or truncate explicit amplitude vectors to the requested subsystem dimension.
- SQR angle arrays are still padded to the storage dimension in `cqed_sim/pulses/calibration.py`.
- The synthesis layer still distinguishes the full Hilbert dimension from restricted subspace dimension in `cqed_sim/unitary_synthesis/subspace.py`.

Direct tests: `tests/test_11_model_invariants.py`, `tests/test_24_displacement_selective_spectroscopy.p` `tests/test_26_state_prep_and_measurement.py`, `tests/test_27_three_mode_model.py`, `tests/unitary_synt

16 Test Suite Audit

File	Targets, type, and main assertion	Status
`tests/test_01_sanity_and_freeu.py`	unit and numerical sanity checks for Hermiticity, free evolution, and Kerr phase accumulation in the two-mode runtime.	Passed
`tests/test_02_cavity_drive_and_kerr.py`	unit and regression checks for storage drive behavior and Kerr response.	Passed
`tests/test_03_dispersive_and_ramsey.py`	dispersive phase scaling and Ramsey pull in the two-mode model.	Passed
`tests/test_04_xy_phase_and_overlap.py`	XY phase convention and overlap cancellation for qubit pulses.	Passed
`tests/test_05_detuning_and_frames.py`	rotating-frame detuning signs and frame handling.	Passed
`tests/test_06_leakage_drag_and_higher_order.py`	multilevel leakage, DRAG behavior, and higher-order term smoke checks.	Passed
`tests/test_07_convergence_regression.py`	truncation convergence and regression against a stored reference.	Passed
`tests/test_08_timeline_and_hardware.py`	waveform timeline assembly and basic hardware distortion semantics.	Passed

File	Targets, type, and main assertion	Status
`tests/test_09_runtime_policy.py`	runtime-budget and fast-path regression checks.	Passed
`tests/test_10_chi_convention.py`	direct checks of runtime χ sign, transition-pull meaning, and observable defaults.	Passed
`tests/test_11_model_invariants.py`	model dimensions, invariants, and expectation consistency.	Passed
`tests/test_12_pulse_semantics.py`	pulse-area and pulse-shape semantics.	Passed
`tests/test_13_hardware_extended.py`	ZOH, low-pass, sideband, quantization, and crosstalk behavior.	Passed
`tests/test_14_dissipation.py`	explicit T_1, T_ϕ, and storage damping checks using collapse operators.	Passed
`tests/test_15_numerics_performance.py`	runtime, compile-cache, and runtime scaling smoke checks.	Passed
`tests/test_16_ideal_primitives_and_extractors.py`	ideal primitives/SNAP/SQR primitives and the two-mode extractor path.	Passed
`tests/test_17_gate_library_integration.py`	gate-library and integration of the library-level gate path.	Passed
`tests/test_18_noise_and_sideband.py`	sideband exchange and combined noise behavior in the two-mode runtime.	Passed
`tests/test_19_fock_tomo.py`	reusable tomography and leakage-calibration helpers.	Passed
`tests/test_20_gaussian_iq_convention.py`	Gaussian I/Q axis, phase, and detuning sign convention.	Passed
`tests/test_21_qubox_convention_reconciliation.py`	reconciliation between runtime conventions and externally documented qubox-style conventions.	Passed
`tests/test_22_support_aware_objective.py`	support-aware objective wiring in synthesis-related helpers.	Passed
`tests/test_23_sqr_additive_amplitude_convention.py`	additive α convention for SQR tones.	Passed
`tests/test_24_displacement_selective_spectroscopy.py`	selective-spectroscopy frequencies and displacement conventions.	Passed
`tests/test_25_tensor_product_conventions.py`	checks of tensor ordering, block indices, and active σ_z sign.	Passed
`tests/test_26_state_prep_and_measurement.py`	measurement ordering, mixed initial states, sampled qubit measurement, and the direct prepare-compile-simulate-measure workflow.	Passed
`tests/test_27_three_mode_model.py`	explicit basis-index, operator-embedding, partial-trace, analytic diagonal-energy, two-mode reduction, and dispersive shift checks for the three-mode model.	Passed
`tests/test_28_three_mode_drives_and_noise.py`	storage/readout ringdown, transmon T_1, transmon dephasing, positivity, and readout-response checks for the three-mode path.	Passed
`tests/analysis/test_spectrum.py`	vacuum-referenced eigenspectrum output, bare-state labeling, sideband level-spacing interpretation, and energy-level plotting smoke checks.	Passed
`tests/test_31_multilevel_sideband_extensions.py`	bare-sideband Hermiticity, $g \leftrightarrow f$ swap, detuned synchronization, shelving isolation, and multilevel-noise regression checks.	Passed
`tests/test_32_kerr_sign_notebook_regression.py`	notebook-scale Kerr evolution matches the documented self-Kerr sign rather than a flipped-sign surrogate.	Passed
`tests/test_33_usage_examples_spectroscopy_sign.py`	fixed-Fock negative-χ spectroscopy regression plus a guard that `examples/workflows/cqed_sim_usage_examples.ipynb` plots physical transition detuning instead of raw carrier.	Passed
`tests/test_34_universal_cqed_model.py`	model-to-model equivalence to the compatibility wrappers, multilevel anharmonicity, and cavity-only runtime support.	Passed
`tests/unitary_synthesis/test_subspace.py`	subspace-basis ordering and round-trip embedding.	Passed
`tests/unitary_synthesis/test_metrics.py`	fidelity gauges and leakage metrics.	Passed
`tests/unitary_synthesis/test_primitives_and_backends.py`	primitives and backends for pulse backends.	Passed

File	Targets, type, and main assertion	Status
`tests/unitary_synthesis/test_optimization.py`	optimization and time optimization.	Passed
`tests/unitary_synthesis/test_synthesis_helpers.py`	synthesis helpers, and MPS-target behavior.	Passed with one optional skip
`tests/unitary_synthesis/test_free_evolve_cond_phase.py`	free-evolve conditional phase formulas.	Passed
`tests/unitary_synthesis/test_hardware_grid_parallel.py`	hardware, grid, parallel, and parallel-multistart behavior.	Passed
`tests/unitary_synthesis/test_progress_reporting.py`	progress reporting and serialization.	Passed
`tests/unitary_synthesis/test_reporting_and_determinism.py`	reporting and deterministic behavior.	Passed
`tests/unitary_synthesis/test_time_policy_and_cond_phase.py`	time-policy and conditional-phase decomposition.	Passed
`examples/audits/tests/test_experiment_contracts_audit2.py`	experiment contracts of audit 2, tensor ordering, waveform signs, and relative phases.	Passed
`examples/workflows/tests/test_square_workflow_artifacts.py`	square workflow artifact serialization and replay.	Passed
`examples/studies/tests/test_snap_opt_convergence.py`	SNAP optimization convergence regression for the SNAP optimization study.	Passed
`examples/studies/tests/test_snap_opt_achieves_zero_coherent_error_above_the_empirical_limit.py`	SNAP optimization achieves zero coherent error above the empirical limit.	Passed
`examples/studies/tests/test_snap_opt_generalization_sweeps.py`	SNAP optimization generalization sweeps, optimizer failure, runtime budget, and parameter sweeps.	Passed
`examples/studies/tests/test_snap_opt_preserves_total_frequency_content.py`	SNAP optimization preserves total frequency content produced.	Passed
`examples/paper_reproductions/tests/test_PRL-133_convergence_and_runtime.py`	PRL-133 convergence and runtime-budget regression.	Passed
`examples/paper_reproductions/tests/test_PRL-133_cross-frequency_control.py`	PRL-133 cross-frequency control test.	Passed
`examples/paper_reproductions/tests/test_PRL-133_optimization_limit.py`	PRL-133 optimization-limit regression.	Passed
`examples/paper_reproductions/tests/test_PRL-133_components_consistency.py`	PRL-133 components consistency.	Passed
`examples/paper_reproductions/tests/test_PRL-133_input_bounds.py`	PRL-133 input bounds and fail-fast validation.	Passed
`examples/paper_reproductions/tests/test_PRL-133_interaction_regression.py`	PRL-133 interaction regression.	Passed
`examples/paper_reproductions/tests/test_PRL-133_unitarity_trace.py`	PRL-133 unitarity and trace-preservation checks.	Passed
`examples/paper_reproductions/tests/test_PRL-136_reproduce_baseline.py`	PRL-136 reproduce baseline smoke test.	Passed
`examples/smoke_tests/tests/test_sanity.py`	sanity and partial-workflow sanity checks.	Passed
`examples/smoke_tests/tests/test_sqr_SQR_calibration.py`	SQR calibration smoke and guarded-benchmark checks.	Passed
`examples/smoke_tests/tests/test_sqr_speed_limit_analytical.py`	speed-limit analytical equivalence checks.	Passed

17 Test Execution Results

Commands used:

```
E:\Programs\python.exe -m pytest tests -q
E:\Programs\python.exe -m pytest examples/audits/tests examples/workflows/tests \
  examples/studies/tests examples/paper_reproductions/tests \
  examples/smoke_tests/tests -q
```

Observed results:

- `tests/`: 222 collected, 221 passed, 1 skipped, 0 failed.
- `examples/*/tests`: 49 collected, 49 passed, 0 failed.
- The single skip is the optional external MPS-reference check in `tests/unitary_synthesis/test_synthesis_an`

- Warning-only notes: two `numpy.trapz` deprecation warnings in `tests/test_12_pulse_semantics.py` and `tests/test_20_gaussian_iq_convention.py`, plus one QuTiP `sqrtm` singular-matrix warning in the open-system sideband fidelity regression.

Interpreter note: `E:\Program Files\Python311\python.exe` from `AGENTS.md` is not present in this workspace, so the suite was run with `E:\Programs\python.exe` (Python 3.11.8).

18 Verified Behaviors Backed by Tests

- Two-mode tensor ordering, basis labels, block indexing, and active Pauli/Bloch sign conventions are directly tested.
- The runtime meaning of χ as a projector-based conditional transition shift is directly tested, including the rule that negative χ lowers the qubit transition with photon number.
- The textbook-to-runtime projector translation is supported by direct σ_z sign tests and by the runtime χ transition tests, even though the exact algebraic translation is not asserted by a single dedicated unit test.
- The new three-mode tensor ordering, operator embeddings, partial traces, diagonal energies, and analytic dispersive shifts are directly tested.
- The three-mode solver path is directly tested for drive isolation, storage/readout-specific damping, transmon relaxation, transmon dephasing, positivity, and qubit-conditioned readout response.
- State preparation helpers are directly tested for ordering and mixed-state behavior.
- The qubit measurement abstraction is directly tested for exact probabilities, confusion-matrix application, finite-shot sampling, and synthetic I/Q output shape.
- The static-spectrum helper is directly tested for vacuum-referenced energies, bare-state labeling, and sideband-relevant energy differences.
- The direct prepare-compile-simulate-measure composition path is directly tested as the recommended library usage pattern.
- The moved example workflows still execute correctly in their new locations.

19 Gaps Between Code Conventions and Test Coverage

- The runtime-to-synthesis translation helper now exposes synthesis coefficients aligned with the runtime convention, but the library still does not force every synthesis-side caller to route through that helper automatically.
- Higher-order χ_2, χ_3, and higher-order Kerr coefficients are covered indirectly, but not with isolated sign-and-factor analytic tests of the same style now used for the three-mode diagonal-energy checks.
- The readout-chain model is physically motivated, but it is still an effective resonator/filter/amplifier abstraction rather than a full microwave network or POVM calibration workflow.
- The backend path is dense and piecewise constant; it is intended for parity and small-system acceleration checks, not as a drop-in replacement for every QuTiP workflow.

20 Potential Ambiguities

Moderate: the repository migrated on March 11, 2026 from an older $-\chi n_c n_q$ convention to the current $+\chi n_c n_q$ convention. Historical notes or saved plots created before that migration may still use the old sign.
Moderate: higher-order pull and Kerr parameters are falling-factorial coefficients, not ordinary polynomial coefficients.
Moderate: the runtime writes transmon physics in terms of n_q rather than $\sigma_z/2$, so ground-state energy offsets differ from some textbook presentations.
Moderate: `EnergySpectrum.energies` are vacuum-referenced in the selected frame, so rotating-frame spectra can contain negative excited-state energies even though the vacuum is fixed at zero.
Moderate: synthetic I/Q output is a convenience sampling layer, not a calibrated hardware-response model.
Low: the explicit complex-envelope sign $e^{+i(\omega t+\phi)}$ is repo-specific and must be preserved when adding pulse families.
Low: `FrameSpec` still stores the storage frame in the legacy field `omega_c_frame`; the `omega_s_frame` alias removes ambiguity but does not rename the underlying field.

21 Summary

`cqed_sim` is now a coherent reusable library for pulse-level cQED simulation with a universal subsystem-based model core plus thin compatibility wrappers for the two-mode transmon-storage path and the three-mode storage-transmon-readout extension. The authoritative runtime conventions are: qubit-first tensor ordering; $|g\rangle = |0\rangle$, $|e\rangle = |1\rangle$; QuTiP-standard Pauli signs; rad/s internally and seconds for time; complex envelopes with $e^{+i(\omega t+\phi)}$ and therefore `carrier` $= -\omega_{\text{transition}}$ for resonant rotating-frame spectroscopy; dispersive physics written in terms of $n_q = b^\dagger b$ with $+\chi n_c n_q$-style conditional terms; and Lindblad collapse operators as implemented in `cqed_sim/sim/noise.py`. Negative χ now means that higher cavity photon number lowers the qubit transition frequency, and positive Kerr coefficients raise adjacent bosonic transition spacings. The experiment-facing layer now has two measurement modes: a default lightweight exact-probability path and an opt-in physical readout-chain path with resonator response, Purcell filtering, amplifier noise, and reduced-state backaction. The library also exposes optional nonlinear/exchange Hamiltonian terms, aligned runtime/synthesis parameter translation helpers, calibration-target utilities, and an opt-in dense NumPy/JAX backend path.